

Physicalized Model Weights - cycles 35-37

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Physicalized Model Weights - cycles 35-37

Abstract

Cycles 35-37 did not add new scientific evidence, executable validation, milestone state, or research artifacts to the physicalized model weights campaign. Instead, they formalized an admission-control boundary: the campaign remains trigger-gated at the validated M-PUBLICBASE-SYNTH-1 endpoint, and no further worker or auditor cycle should be opened unless a validated trigger appears.

The supplied audit report for this range made the controlling decision explicit: PIVOT. This was not a pivot away from the campaign’s prior conclusions. It was a pivot away from treating no-op admission handling, watch-state reconfirmation, or trigger-gate restatement as campaign progress.

The validated endpoint from earlier cycles remains unchanged. The Phase 2 downgrade is preserved, the Phase 4 reopen condition remains unchanged, public benchmark evidence remains programmable-baseline context only, and the endpoint counters remain zero or false:

Endpoint field	Current value
current_superiority_claim_count	0
actual_reopen_candidate_count	0
new_reopen_gate_count	0
current_artifacts_reopen	false

Introduction

The long-exposure campaign investigates whether useful parts of neural-network inference can be “physicalized” into hardware: fixed, semi-fixed, or physically encoded substrates that reduce repeated software execution and memory movement. Earlier validated work narrowed the conclusion substantially. Full fixed frontier-model physicalization remains rejected under

current evidence, and the surviving value of the project is a bounded architecture, failure-mode, verification, and evidence-gating study.

The latest substantive validated milestone before this report is M-PUBLICBASE-SYNTH-1. That milestone integrated public programmable-baseline refresh work into the canonical campaign record. It used public MLPerf Inference context, including MLCommons documentation and public v6.0 result records, to strengthen or preserve the programmable-baseline null rather than reopen physicalized superiority [10]-[13]. Vendor benchmark material remained secondary context only [14].

Cycles 35-37 occurred after that synthesis. Their purpose was not to build a new model or rerun a benchmark. Their purpose was to decide whether a new cycle was admissible. In this report, “admission control” means the gate that determines whether a new research cycle should be opened at all. A cycle is admissible only if it has a concrete trigger-backed scope: new evidence, changed tool capability, a materially new public-data mapping question, or a nonduplicative handoff artifact requirement.

Approach

The reporter gathered and consolidated the provided cycle records, the supplied audit report, the current campaign manifest, the plan of record, the ledger tail, the reference list, the workspace figure inventory, and the current public-baseline synthesis endpoint summary.

The relevant session records were:

Cycle	Role	Session ID	Role in timeline
35	Researcher	e926d3f3-afaa-4791-91b5-1236adbf1053	Issued a trigger-gated brief and assigned no worker build.
35	Worker	7f35edfb-af67-4e9b-a02f-2f750bf10a98	Built nothing, ran nothing, and reported no trigger.
35	Auditor	40f2437d-03ac-46fc-8b38-3a45412be8cc	Confirmed endpoint counters and issued PIVOT because the cycle was not substantive progress.
36	Researcher	5dde2698-f2be-4989-8ed6-cbf0795751ac	Reframed the state as admission control and said no worker cycle should be instantiated.
36	Worker	22c675b2-8cff-4134-a371-3a1974c0f786	Built nothing and ran nothing because no handoff was issued.

Cycle	Role	Session ID	Role in timeline
36	Auditor	93f77468-7f95-4bb8-9e29-0adc568457f0	Issued PIVOT and directed the campaign not to emit another no-trigger worker brief.
37	Researcher	83f4ccc7-82cb-4b69-9d4b-9615ab99158b	Emitted <code>ADMISSION_BLOCKED_NO_CYCLE</code> .
37	Worker	b7f70785-379b-4a34-91fc-386fc2d4e2c4	Confirmed no worker cycle was opened and no work was run.
37	Auditor	d35d2e50-93f7-40ef-9725-bc99f6a9002d	Issued the final PIVOT for the range: no-cycle admission handling is not campaign progress.

No new figures were produced for this cycle range. Existing workspace figures belong to earlier milestones.

Findings

Cycle 35: Trigger-Gated Boundary

Cycle 35 began with the researcher explicitly stating that the campaign remained at the validated `M-PUBLICBASE-SYNTH-1` endpoint. The researcher checked the four admissible trigger classes and found all absent:

- no lifecycle-valid measured production, shadow, or canary hybrid evidence;
- no relevant compiled-HDL capability change for `M-PROTO-1`;
- no materially new primary public-data mapping scope beyond `M-PUBLICBASE-1`, `M-PUBLICBASE-2`, or `M-PUBLICBASE-SYNTH-1`;
- no nonduplicative handoff artifact class.

The worker session for cycle 35 built nothing and ran no commands. It reported that no files, artifacts, milestones, or ledger events were created.

The auditor agreed that the trigger-gated conclusion was correct but issued `PIVOT`, not `VALIDATED`. The rationale was that the cycle produced no evidence, mechanism update, executable validation, or artifact state. The auditor did run endpoint checks and reported `promise_check` passing with events: 89, plan milestones: 33, with only known orphan report warnings. It also reported `org_check` passing with only known root-file warnings.

Cycle 36: Admission Control

Cycle 36 tightened the framing. The researcher stated that there was no active research sub-topic and that no worker or auditor research cycle should be instantiated without a validated trigger.

The worker response matched that brief: no handoff was issued, no files were edited, no artifacts were created, no validators were rerun, and no ledger event was added.

The auditor again issued `PIVOT`. The decision did not change the endpoint. It clarified process discipline: an admission-control response is not a substantive research cycle. The auditor's guidance was to stop emitting no-trigger worker briefs and to treat the appropriate response as admission-blocked/no cycle when no trigger is present.

Cycle 37: Admission Blocked, No Cycle

Cycle 37 implemented the auditor's guidance directly. The researcher output was the explicit state:

`ADMISSION_BLOCKED_NO_CYCLE`

The researcher did not open a worker cycle, milestone, artifact, validation run, or watch-state reconfirmation. The worker session recorded that admission was blocked and that nothing was built or run. The auditor then issued the range-controlling `PIVOT` decision.

The supplied audit report for this reporting task matches the cycle 37 auditor record. It states that no commands were run, no validators were rerun, no files were edited, no artifacts were generated, and no ledger event was added. It also states that this is the correct behavior for an admission-blocked no-cycle state.

Discussion

The important result of cycles 35-37 is procedural, not scientific. The campaign did not discover new physicalized-weight evidence. It established that continuing to instantiate worker and auditor sessions solely to restate the trigger-gated endpoint creates no campaign progress.

The controlling scientific state remains the one validated at `M-PUBLICBASE-SYNTH-1`:

- Public MLPerf benchmark evidence can update the programmable-baseline prior B , but it does not supply a measured hybrid total H .
- The Phase 2 downgrade remains preserved.
- The Phase 4 reopen condition remains unchanged.
- No current artifact is measured production, shadow, or canary hybrid evidence.
- No current artifact creates a superiority claim, actual reopen candidate, or new reopen gate.

Here, B means the measured best programmable baseline under the same workload accounting. H means the measured hybrid physicalized system total under that same accounting. The campaign's future reopen rule still requires a lifecycle-valid evidence package and an uncertainty-aware win, not public benchmark context alone.

The admissible future triggers remain:

- lifecycle-valid measured production, shadow, or canary hybrid evidence;
- changed compiled-Verilator capability relevant to `M-PROTO-1`;
- materially new primary public-data mapping scope;
- nonduplicative handoff artifact requirement.

If none of those appears, the correct output is admission blocked/no cycle, not another watch-state validation.

Open Questions

No new scientific open questions were created in cycles 35-37.

The standing future questions are unchanged:

- Can a real measured hybrid evidence package pass the full Phase 4 lifecycle, provenance, privacy, workload, baseline, and uncertainty gates?
- If `make` and a C++ compiler become available, can compiled Verilator simulation strengthen prototype verification for M-PROTO-1?
- Will a future primary public-data source create a materially different mapping question not already covered by the public-baseline recency, prior-refresh, and synthesis milestones?
- Will a genuinely new handoff artifact class become necessary without duplicating closure, archive, invariant, deferral, or public-baseline synthesis work?

References

[10] MLCommons, “MLPerf Inference: Datacenter benchmark documentation,” MLCommons. <https://docs.mlcommons.org/inference/>

[11] MLCommons, “MLPerf Inference v6.0 Results,” MLCommons, 2026. <https://mlcommons.org/2026/04/mlperf-inference-v6-0-results/>

[12] MLCommons, “MLPerf Inference v5.1 Results,” MLCommons, 2025. <https://mlcommons.org/2025/09/mlperf-inference-v5-1-results/>

[13] MLCommons, “MLPerf Inference Results v6.0,” GitHub, 2026. https://github.com/mlcommons/inference-results_v6.0

[14] NVIDIA, “MLPerf AI Benchmarks,” NVIDIA. <https://www.nvidia.com/en-us/data-center/resources/mlperf-benchmarks/>

Appendix: Implementation Details

Code Organization

No new scripts, tests, HDL files, documents, data files, or figures were created during cycles 35-37.

The manifest remains a snapshot of the existing campaign artifacts. It was updated only to label the active terminal state as “trigger-gated admission control/no-cycle state.” The cumulative counts did not change:

Category	Count
Authored research scripts	35
Authored research script lines	15,311
Authored tests	32
Authored test lines	4,439
Authored HDL/support files	4
Authored HDL/support lines	241
Authored research docs and diagram sources	34
Authored doc/source lines	2,150
Ledger events	89
Plan milestones	33

Test Results

No cycle 37 validators were run by the researcher, worker, or auditor because the correct state was `ADMISSION_BLOCKED_NO_CYCLE`.

During report preparation, the reporter ran manifest and workspace sanity checks after the manifest wording update:

Check	Result
<code>python3 -m long_exposure.tools.promise_check .</code>	Exit 0; events: 89, plan milestones: 33; known orphan report warnings only.
<code>python3 -m long_exposure.tools.org_check .</code> Manifest sanity check	Exit 0; known root-file warnings only. 193 lines; no ## Key Files section; expected counts preserved; admission-control wording present.

These checks were used to confirm the report's implementation snapshot, not to re-audit the underlying research endpoint.

File and Figure Inventory

No cycle 35-37 figure exists. Existing figures in `physicalized-weights/data/` and `physicalized-weights/docs/` belong to earlier milestones.

Existing rendered cycle reports in `reports/cycles/` now include reports through `report_cycles_32-34.md` and `report_cycles_32-34.pdf`. The current report is intended for `report_cycles_35-37`.

Session References

Source	Session ID
Cycle 35 researcher	e926d3f3-afaa-4791-91b5-1236adbf1053
Cycle 35 worker	7f35edfb-af67-4e9b-a02f-2f750bf10a98
Cycle 35 auditor	40f2437d-03ac-46fc-8b38-3a45412be8cc
Cycle 36 researcher	5dde2698-f2be-4989-8ed6-cbf0795751ac
Cycle 36 worker	22c675b2-8cff-4134-a371-3a1974c0f786
Cycle 36 auditor	93f77468-7f95-4bb8-9e29-0adc568457f0
Cycle 37 researcher	83f4ccc7-82cb-4b69-9d4b-9615ab99158b
Cycle 37 worker	b7f70785-379b-4a34-91fc-386fc2d4e2c4
Cycle 37 auditor	d35d2e50-93f7-40ef-9725-bc99f6a9002d

Cross-Reference Map

Origin	Consuming state	Meaning
<code>public_baseline_synthesis_summary.json</code>	cycles 35-37 admission control	Supplies the controlling endpoint counters and M-PUBLICBASE-SYNTH-1 status.

Origin	Consuming state	Meaning
Cycle 35 auditor session	Cycle 36 researcher brief	Converts trigger-gate reconfirmation from a null-cycle into an admission-control problem.
Cycle 36 auditor session	Cycle 37 researcher brief	Directs the next response to be admission blocked/no cycle when no trigger is present.
Cycle 37 researcher, worker, and auditor sessions	This report	Establishes the final reporting conclusion: no validated trigger, no worker cycle, no artifact, and PIVOT.
MANIFEST.md	Future researcher/worker turns	Records the current active state as trigger-gated admission control/no-cycle with unchanged artifact counts.